



**FACULTY OF INFORMATION AND COMMUNICATION
TECHNOLOGIES**

SUMMER 2025

CONTINUOUS ASSESSMENT

COURSE TITLE:	ARTIFICIAL INTELLIGENCE
COURSE CODE:	CS-4160
INSTRUCTOR:	Dr. Nkandeu Pascal
DATE:	Summer 2025
DURATION:	1hr

Question 1: Real-World AI Deployment Tasks (40 marks)

You are consulting for a logistics company that wants to optimize sales prediction, customer segmentation, route optimization, and chatbot support. Use the data and techniques to build solutions.

a) Multiple Linear Regression (5 marks)

You are given data to predict **Delivery Time (Y)** based on **Distance (X1)** and **Package Weight (X2)**:

Obs	Distance (km)	Weight (kg)	Delivery Time (hrs)
1	5	2	1.5
2	10	3	2.3
3	15	4	3.1
4	20	5	4.0

- Build the regression model
- Predict time for Distance = 12 km, Weight = 3.5 kg
- Interpret coefficients

b) Decision Tree Entropy & Split (3 marks)

A simplified dataset is used for routing method classification:

Weather	Vehicle	Route Type
Rainy	Car	Long
Sunny	Bike	Short
Cloudy	Car	Long
Sunny	Car	Short
Rainy	Bike	Long

- Calculate entropy of Route Type
- Compute Information Gain for splitting on "Vehicle"

c) DBSCAN Clustering (3 marks)

Apply DBSCAN to the data ($\epsilon = 2$, MinPts = 2):

Point	X	Y
A	1	1
B	1	2
C	2	2
D	8	8
E	9	9
F	15	15

- Identify core points
- List cluster assignments and noise points

d) k-NN Classification (5 marks)

You want to classify a new transaction with features (Age=28, Income=55):

ID	Age	Income	Class
A	25	50	Low
B	30	60	High
C	24	45	Low
D	31	58	High

- Use $k=3$ and Euclidean distance
- Determine predicted class
- Visualize decision boundary assumption

e) k-Means Clustering (4 marks)

Initial centroids: $C1 = (2, 3)$, $C2 = (8, 7)$

Point	X	Y
A	1	2

B	2	2
C	9	6
D	8	8

Tasks:

1. **Assign points** to nearest centroid using Euclidean distance. *(2 marks)*
2. **Recalculate centroids** based on new clusters. *(2 marks)*
3. **Repeat one more iteration** if needed. *(2 marks)*

Metrics calculation – Compute **Within-Cluster Sum of Squares (WCSS)** for the final